PARTICULATE MATTER COMPLIANCE REPORT

JEFFREY ENERGY CENTER – #4 DUST COLLECTOR

SSID: 1490001

PREPARED FOR:

EVERGY
JEFFREY ENERGY CENTER
25905 JEFFREY ROAD
ST MARYS, KANSAS 66536

PREPARED BY:

EVERGY TESTING & ANALYTICS
TOPEKA, KANSAS 66612

ET&A PROJECT 25-003

TEST DATE: 2/1/2025

REPORT DATE: 2/28/2025

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TEST REPORT CERTIFICATION

This report, testing details, and approach have been developed under the supervision (including review) of the persons named below. Results contained in this report relate only to the sources tested and the parameters included in the test program.

Evergy Testing & Analytics (ET&A) operates as an air emission testing body (AETB) under a quality management system in conformance with ASTM D7036-04 "Standard Practice for Competence of Air Emission Testing Bodies".

Date: 2/26/2024 Signature________________________

Brandon Jones, QSTI
Manager, Environmental Testing & Analytics
Evergy Testing & Analytics

Qualified Individual Statement

I, Brandon Jones, hereby agree on this day April 9, 2024, that as a qualified source testing individual, all projects conducted under my supervision will conform to the policies and procedures outlined in ASTM D 7036-04, 40CFR60, 40CFR75, all Federal, State, and local agencies, and the Evergy Testing & Analytics Quality Manual.

(Signature of Individual Signing Statement)

4/9/2024
(Date of Signature)

4/9/2024
(Date of QI Certification)

V
(Group of Certification)

(Signature of Supervisor/Witness)

4/9/2024
(Date of Signature)

INTRODUCTION

Facility Name and Location:
Jeffrey Energy Center
25905 Jeffrey Road
St Marys, KS 66536

Reason for Conducting Test:
To conduct the initial Particulate Matter (PM) Compliance testing at Jeffrey Energy Center, #4 Dust Collector, as required by Part 60 Subpart Y.

Test Date(s):
February 1, 2025

Unit description:
#4 Dust Collector is a dust collection system which pulls air from above conveyor belts directly receiving coal from railcars.

Testing Organization:
Evergy
Evergy Testing & Analytics
818 Kansas Ave
Topeka, KS 66612

Personnel Conducting Test:
Brandon Jones (QSTI: V) – Manager, Environmental Testing & Analytics
Jeremy Duis – Sr. Manager of CEMS and Reporting

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TEST RESULTS

EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data

	Run #1	Run #2	Run #3	Averages	Permit Limit	
Date:	2/1/2025	2/1/2025	2/1/2025			
Start time:	10:19	14:04	15:20			
Stop time:	13:05	15:09	16:24			
Filterable PM (gr/dscf):	0.0019	0.0020	0.0028	0.0023	0.010	PASS

TESTING METHODOLOGY

The following section describes the procedures used for the Particulate Matter (PM) compliance testing. The test was conducted in such a way that it yielded measurements representative of the emissions from the source. The PM testing was performed to demonstrate compliance per Title 40, Code of Federal Regulations (CFR), Part 60, Subpart Y. The PM compliance testing consisted of sampling for concentrations of particulate matter in the flue stream at the #4 Dust Collector exhaust stack.

A minimum of three (3) test runs for particulate matter emissions must be performed while the coal from the train is being dumped continuously.

REFERENCE METHOD AND RELATIVE ACCURACY PROCEDURES

Evergy Testing & Analytics personnel tested in accordance with the following U.S. EPA source emissions test methods referenced in *40 CFR 60, Subpart Y.*

- Method 1 – Sample and Velocity Traverses for Stationary Sources.
- Method 2 – Determination of Velocity and Volumetric Flow Rate in Stationary Sources
- Method 3A – Determination of Oxygen and Carbon Dioxide Concentrations in Emissions from Stationary Sources (Instrumental Analyzer Procedure)
- Method 4 – Determination of Moisture in Emissions from Stationary Sources
- Method 5 – Determination of Particulate Matter Emissions from Stationary Sources

SAMPLING AND ANALYTICAL PROCEDURES

USEPA Method 1 was used to determine the appropriateness of the existing ports as the sampling location and to determine the number of sampling points to be used when sampling at this location.

Upstream Duct Diameters	Downstream Duct Diameters	Number of Ports	Number of Point per Port	Distance from Far Wall to outside of Port	Port length	Depth of Duct
3.3	1.5	2	6	52.0	4.25	48.0

After determining the number of sampling points, a Type “S” pitot tube such as the one defined in USEPA Method 2 was used to determine if cyclonic flow conditions existed at the sampling location. ET&A leveled and zeroed a manometer and connected a Type “S” pitot tube and leak-checked the system. The pitot tube was positioned at each traverse point, in succession, so that the planes of the face openings of the pitot tube were perpendicular to the stack cross-sectional plane. This position was referred to as "0° reference point." The differential pressure (ΔP) reading at each traverse point was recorded. The pitot tube was rotated until a null reading was obtained. The value of the rotation angle was determined to the nearest degree using an angle finder. After this technique was applied at each traverse point, the average of the absolute value of the angles was determined. The average angles were $\leq$ 20 degrees and the overall flow conditions were found to be acceptable.

After determining the number of sampling points to use, USEPA Method 2 was used to perform preliminary velocity measurements of the flue gas in the stack while checking the system for flow conditions. Preliminary velocity measurements were made in each stack to determine the appropriate nozzle size to use to maintain isokinetic sampling rates during the sampling. Velocity measurements were also made while collecting the samples to ensure the samples were collected isokinetically and to determine the pollutant concentrations and mass emission rates.

The stack gas velocity was determined from the measurement of the velocity head at each traverse point, with a calibrated Type "S" pitot tube connected to an inclined manometer. Measurement of the stack gas temperature was made with a calibrated Type "K" thermocouple, and determination of the stack gas composition was made using USEPA Method 3A.

ET&A made measurements of the face opening alignments, external tubing diameter, and base-to-opening plane distances of the pitot tubes. These measurements meet the design criteria in USEPA Method 2 for a Type “S” pitot tube and, therefore, a baseline coefficient value of 0.84 was assigned to the pitot tube. Verification of these measurements was recorded onto a pitot tube calibration worksheet and presented as part of the final report. Once the measurements of the pitot tube were completed, the system was assembled and leak-checked prior to making any measurements.

In addition to the velocity head measurements, USEPA Method 3A was used to determine O_2 and CO_2 concentrations in the flue gas. The O_2 and CO_2 concentrations were made to calculate the flue gas molecular weight. The latter measurements were also made to complete the preliminary velocity calculations necessary to establish the sampling conditions for setting up the actual pollutant measurement test runs.

The Method 3A sample was extracted from the stack through a heated stainless-steel probe into a chilled moisture knockout system using ¼ -inch PTFE sample line and a regulated K&F single diaphragm sample pump. The sample was then stored in a Tedlar Bag for analysis. The bag sample was then analyzed for a set amount of time in order to record a stable reading. The analyzer output was recorded continuously by a RATAView Data Acquisition System (DAS) manufactured by ESCSpectrum.

USEPA Protocol-1 calibration gases were used to calibrate the analyzer system. A calibration error test was performed on each analyzer by introducing a zero, mid and high-level calibration gas directly to the analyzer inlet. A zero gas (nitrogen) of less than 0.25 percent of span, a certified Protocol-1 mid-level (40 to 60 percent) of span and a certified Protocol-1 high-level (100 percent) of span were used for the analyzer calibration error test.

The measurement system was required to meet the following QA/QC criteria for this test program:

- Analyzer Calibration Error: ± 2.0 percent of the calibration span for the zero, mid and high-level calibration gases.
- System Bias: ± 5.0 percent of the analyzer calibration span for the zero and upscale calibration gases.
- Calibration Drift: ± 3.0 percent of the analyzer calibration span for the zero and mid or high-level gases.

REFERENCE METHOD CEMS CALIBRATION GASSES

Pollutant	Level	Cylinder #	Certified Value
CO_2 / O_2	Zero	UHP Nitrogen	Zero
	Mid	CC199274	9.61 / 9.98
	High	DT0031874	20.10 / 20.00

ANALYZER INFORMATION AND TECHNOLOGY

Brand	Model	Compounds Measured	Measurement Technology	Range	Sample Condition
Teledyne	T802	CO_2/O_2	NDIR / Paramagnetic	0-25 %	Cold, Dry

For this testing, the Method 4 sampling train was incorporated into the Method 5 sampling trains. The moisture content of the sample gas stream was determined by extracting the gas sample at a known and regulated rate through the glass condenser train maintained below 68 degrees Fahrenheit (°F) in an ice bath with a vacuum pump. The volume of gas sampled was measured with a calibrated dry gas meter. The volume of moisture collected was measured gravimetrically with an analytical balance with a tolerance of 0.1 gram (g).

METHOD 5

The sampling and analytical procedures used during the emission testing for PM were conducted in accordance with the procedures outlined in USEPA Method 5. The probe and filter exit temperature was maintained at 250 ± 25 °F, in accordance with the requirements of the test method. The sampling equipment consisted of the following equipment and components:

1) **Probe assembly**
   a) Nozzle – Calibrated borosilicate glass with a sharp, tapered leading edge.
   b) Probe - Stainless steel sheath with a 5/8-inch diameter glass insert wrapped with nichrome wire; electronically controlled and capable of maintaining 250 ± 25 ºF with a calibrated Type "K" thermocouple.
   c) Pitot - Calibrated stainless-steel Type "S" pitot tube constructed and attached to the probe according to the specifications outlined in USEPA Method 2, 40 CFR 60.
   d) Thermocouple - Calibrated Type "K" thermocouple attached to the sample probe such that the tip has no contact with any metal and does not interfere with the pitot tube face openings and the sample nozzle.
2) **Filter Box Assembly**
   a) Filter Holder - Borosilicate glass, with a Teflon frit filter support and a silicone rubber gasket.
   b) Filter Heating System – The heating system is capable of monitoring and maintaining the temperature around the filter to ensure the sample gas temperature exiting the filter is 320 ± 25 ºF during sampling.
   c) Filter Temperature Sensor – A thermocouple capable of measuring temperature to within ±3 °C (5.4 °F) is installed so that the sensing tip of the temperature sensor the temperature of the filter. The filter temperature sensor was monitored and recorded during sampling to ensure a sample gas temperature exiting the filter of 250 ± 25 °F.
3) **Impingers**
   a) The back-half Method 5 impingers consisted of four sequential impingers. Impingers one and two contained 100 ml each of DI water. Impinger three was empty to prevent liquid carry over. The final impinger contained at least 300 grams of silica gel.
   b) The first and third impingers were of the Modified Greenburg-Smith design. The second impinger was of the Greenburg-Smith design with a standard orifice tip. Final gas exit temperature was measured with a calibrated Type "K" thermocouple immersed in the gas stream.
4) **Control Box**
   a) The control box is a module containing a vacuum gauge, external leak-free pump, thermocouples capable of measuring temperature within ± 5°F, dry gas meter with a minimum of 2 percent accuracy, valves and related equipment to determine sample gas volume.

The control box was leak-checked from pump to orifice at a minimum of 15 inches of mercury (in. of Hg) vacuum. The selected nozzle was attached to the sample probe. The moisture sample train was prepared per Method 4. The cold box portion was packed with crushed ice to maintain final gas exit temperature below 68 ºF.

A pre-weighed quartz fiber filter, prepared per Method 5, was placed inside the filter holder onto a Teflon® support frit. The filter holder was positioned into the filter assembly unit and connected to the sample probe and impinger train with interconnecting glassware. The entire assembled sample system was leak-checked by plugging the inlet to the probe nozzle and pulling a vacuum of at least 15 inches of Hg. A leak rate less than 0.02 cubic feet per minute (cfm) was determined and considered acceptable as outlined in 40 CFR 60. The pitot tube system was also vacuum and pressure leak-checked at greater than 3 inches of water column. The traverse sampling points were marked on the probe for easy visibility.

Temperature controllers for the probe were set to maintain a temperature of 250 ±25°F. The nozzle was placed at the first traverse point and positioned directly facing the stack gas stream. The sampling pump was started and the meter set to the appropriate sampling rate at each of the individual traverse points. At the conclusion of each test run, the sample pump was turned off, the probe was removed from the stack, and the final sample train measurements were recorded. A final leak-check of the system was performed at the conclusion of every test run, as previously described at the highest vacuum encountered during testing. A leak-check of the pitot system was performed as previously described. The final leak-checks were determined to be within the guidelines outlined in Method 5. All velocity measurements, sample train measurements, moisture data, and leak-check measurement were recorded onto the field data worksheets.

The PM filter holder and the interconnecting hardware were removed from the sample probe and the openings sealed. Likewise, the sample probe was disconnected. The impinger train, sample probe, PM filter holder, and interconnecting glassware were relocated to the sample collection area.
The PM filter holder was labeled with sample identification and prepared for transfer to the ET&A's on-site laboratory. The nozzle was carefully removed from the sample probe and rinsed with acetone into a glass container. Acetone was used to rinse the probe liner, ferrules, nozzle, and all interconnecting glassware utilized between the sample nozzle and filter holder. A nylon brush was used to loosen any adhering PM and subsequent washes and rinses were collected into the sample container. Acetone was also used to rinse any PM adhering to the nylon brush into the sample collection container.
An acetone blank sample was collected at the sample cleanup area at the facility. The blank sample was analyzed using the same procedures as the test samples. The acetone blank was found to be within specifications and did not impact the results.

Prior to conducting the final analysis, a preliminary analysis was conducted in the field following sample recovery to give real time performance indicators to facility personnel. Upon arrival at the ET&A on-site mobile laboratory, the integrity of the PM filter holders was inspected and found to be intact. The quartz fiber filter was removed from the filter holder and placed into the original petri dish. Any excess PM contained within the front half of the filter holder was brushed into the petri dish. The front half of the filter holder was rinsed and cleaned with the nylon brush and the washes were collected into the appropriate sample container with the other sample train washes for each respective test run.
The preliminary particulate analysis was conducted by ET&A in accordance with USEPA Method 5 with shortened desiccation periods and intervals between weights. The filters were transferred into the desiccator for an hour to obtain a weight and then weighed every 30 minutes, minimum, until two consecutive weights were obtained that agreed within ± 0.5 milligrams.

The acetone rinses and an acetone blank were evaporated in tared beakers on a hot plate to determine the level of particulate collected in the nozzle, probe liner and the front half of the filter holder and to determine the amount of residual weight each beaker retained due to acetone impurities. After evaporating to dryness, the beakers were transferred to a desiccator for one hour to obtain a weight and then weighed every 30 minutes, minimum, until two consecutive weights were obtained that agreed within ± 0.5 milligrams. Each filter, acetone washing, and acetone blank was individually weighed on an analytical balance with a sensitivity of 0.1 milligram.

The final particulate analysis was conducted by ET&A in accordance with USEPA Method 5. The filters and the beakers containing the particulate recovered from the probe and nozzle rinses were transferred into the desiccator for twenty-four hours to obtain a weight and then weighed every six hours, minimum, until two consecutive weights were obtained that agreed within ± 0.5 milligrams.

QUALITY ASSURANCE / QUALITY CONTROL SUMMARY

ET&A has established quality assurance and quality control (QA/QC) guidelines for providing quality sampling and analytical data from source tests. These QA/QC procedures were implemented to ensure the acceptability and reliability of the data generated. In summary, an appropriate degree of data quality was maintained throughout this project. The sampling trains were leak checked prior to and immediately after sampling. Leak rates for the isokinetic sampling trains were less than the maximum criterion of 0.02 cubic feet per minute.

The isokinetic sampling rates for the isokinetic sampling trains were within the 100% ± 10% criterion established for isokinetic sampling. The pitot leak checks were stable for at least 15 seconds. USEPA Method 5 requires the calibration of the metering system after each field use. This is typically performed in a laboratory environment. If it fails the test, the test results are invalidated. This makes a field calibration procedure highly desirable. USEPA's approved alternative method (ALT-009) based on the pre-test meter coefficient found in Method 5 Section 4.4.1 has been approved as a post-test calibration procedure. The procedure states that, "If the average of the three runs is within ± 5% of the y-value, the Method 5 sampling system passes the post calibration requirements."

DATA UNCERTAINTY

Both qualitative and quantitative factors contribute to field measurement uncertainty and should be taken into consideration when interpreting the results presented in this test report. There are several factors that can affect qualitative and quantitative measurements.

Qualitative uncertainty factors include, but are not limited to, unknown chemical interferences, sample matrix interactions, environmental conditions, sample handling and instrument operation and maintenance. To reduce the impact of these qualitative uncertainty factors, ET&A has developed a set of Standard Operating Procedures (SOPs) in accordance with our corporate quality assurance guidelines and ASTM D 7036-04.

Quantitative uncertainty factors known to directly affect uncertainty include the accuracy of calibration standards as well as the precision and accuracy of instrument measurements and the test methods utilized. To reduce the impact of these quantitative uncertainty factors, ET&A utilizes testing and analytical methodology that has been approved by EPA or the American Society for Testing and Materials (ASTM) where applicable. In addition, ET&A personnel perform routine instrument and equipment calibrations according to manufacturer's guidelines and/or test method specifications.

The limitations of the various methods, instruments, equipment, and materials utilized during this project have been reasonably considered to be in accordance with the project data quality objective, but the ultimate impact of the cumulative uncertainty of this project is not fully identified within the results of this test report.

COMPLIANCE DETERMINATION

EPA Method 5 : Determination of Particulate Emissions from Stationary Sources

Field Data

	Run #1	Run #2	Run #3	Averages
Date:	2/1/2025	2/1/2025	2/1/2025	
Start time:	10:19	14:04	15:20	
Stop time:	13:05	15:09	16:24	
Sample volume (ft^3):	59.258	57.417	60.758	59.144
Sampling time (minutes):	60	60	60	60
Stack temp. (^oF):	48	50	50	49.2
Meter temp. (^oF):	74	83	90	82.0
Barometric pressure (in Hg):	28.74	28.74	28.74	28.74
Static pressure (in H_2O):	-8.34	-8.34	-8.34	-8.340
Stack pressure (in H2O):	28.13	28.13	28.13	28.127
Oxygen (%dry):	20.8	20.8	20.8	20.79
Carbon Dioxide (%dry):	0.0	0.0	0.0	0.00
Moisture (gm):	8.8	8.6	9.8	9.07
ΔH (in H_2O):	2.992	2.800	3.017	2.936
$\sqrt{\Delta p}$ (in H_2O):	0.5361	0.5299	0.5330	0.533
DGM Y_c:	0.986	0.986	0.986	0.986
Pitot Tube Coefficient (C_p):	0.84	0.84	0.84	0.840
Nozzle diameter (inches):	0.308	0.303	0.308	0.306
Stack diameter (inches):	48.00	48.00	48.00	48.00

Method 5 Laboratory Data

	Run #1	Run #2	Run #3	Averages
Beaker Tare Weight (g):	51.7665	51.8116	53.2783	
Beaker Final Weight (g):	51.7732	51.8184	53.2877	
Filterable PM mass (probe) (g):	0.0067	0.0068	0.0094	0.00763
Filter Tare Weight (g):	0.3524	0.3528	0.3525	
Filter Final Weight (g):	0.3527	0.3531	0.3532	
Filterable PM mass (filter) (g):	0.0003	0.0003	0.0007	0.00043
Acetone Wash (ml):	56	59	60	

Blank Volume (ml)	200
Reagent Blank (g)	0.00010
Acetone Density g/l	0.7846
Max Ca%residue	0.0010%
Ca%residue	0.0001%

Calculations

	Run #1	Run #2	Run #3	Averages
Maximum allowed blank correction (g):	0.00044	0.00046	0.00047	0.00046
Acetone blank PM, if allowed (g):	0.00003	0.00003	0.00003	0.00003
Blank corrected PM (g):	0.00697	0.00707	0.01007	0.00804
Sample volume (dscf) (Vmstd):	55.953	53.281	55.711	54.9819
Moisture volume (scf) (Vwstd):	0.415	0.405	0.462	0.4275
Moisture Fraction Measaured (BWSmsd):	0.007	0.008	0.008	0.008
Moisture Fraction @ Saturation (BWSsat):	0.012	0.013	0.013	0.012
Moisture Fraction (BWS):	0.007	0.008	0.008	0.008
Molecular weight (dry) (Md):	28.83	28.83	28.83	28.832
Molecular weight (actual/wet) (Ms):	28.75	28.75	28.74	28.748
Gas velocity (ft/sec) (Vs):	30.5	30.2	30.4	30.37
Stack area ft^2 (As):	12.57	12.57	12.57	12.57
Gas flow (acfm) (Qa):	23,007	22,773	22,922	22900.5
Gas flow (dscfm) (Qs):	22,314	22,017	22,126	22152.3
% Isokinetic:	101.5	101.2	101.9	101.5

	Run #1	Run #2	Run #3	Averages	Permit Limit	
Filterable PM (gr/dscf):	0.0019	0.0020	0.0028	0.0023	0.010	PASS

EXAMPLE CALCULATIONS

Plant/Unit:	**#4 Dust Collector**
Project No.:	**25-003**
Run No.:	**1**
Test Type:	**M5**

Meter Pressure (Pm), in. Hg

$$Pm = Pb + \frac{\Delta H}{13.6}$$

where,

Pb	28.74	= barometric pressure, in. Hg
ΔH	2.992	= pressure differential of orifice, in H_2O
Pm	28.96	= in. Hg

Absolute Stack Gas Pressure (Ps), in. Hg

$$Ps = Pb + \frac{Pg}{13.6}$$

where,

Pb	28.74	= barometric pressure, in. Hg
Pg	-8.34	= static pressure, in. H_2O
Ps	28.13	= in. Hg

Standard Meter Volume (Vmstd), dscf

$$Vmstd = \frac{17.647 \times Y \times Vm \times Pm}{Tm}$$

where,

Y	0.986	= meter correction factor
Vm	59.258	= meter volume, cf
Pm	28.96	= absolute meter pressure, in. Hg
Tm	533.7	= absolute meter temperature, °R
Vmstd	55.953	= dscf

Standard Wet Volume (Vwstd), scf

$$Vwstd = 0.04707 \times Vlc$$

where,

Vlc	8.8	= volume of H_2O collected, ml
Vwstd	0.415	= scf

Moisture Fraction (BWSsat), dimensionless (theoretical at saturated conditions)

$$BWSsat = \frac{10^{6.37 - \left(\frac{2,827}{Ts + 365}\right)}}{Ps}$$

where,

Ts	48.0	= stack temperature, °F
Ps	28.13	= absolute stack gas pressure, in. Hg
BWSsat	0.012	= dimensionless

Moisture Fraction (BWS), dimensionless (measured)

$$BWS = \frac{Vwstd}{(Vwstd + Vmstd)}$$

where,

Vwstd	0.415	= standard wet volume, scf
Vmstd	55.953	= standard meter volume, dscf
BWS	0.007	= dimensionless

Moisture Fraction (BWS), dimensionless

$$BWS = BWSmsd \text{ unless } BWSsat < BWSmsd$$

where,

BWSsat	0.012	= moisture fraction (theoretical at saturated conditions)
BWSmsd	0.007	= moisture fraction (measured)
BWS	0.007	

Molecular Weight (DRY) (Md), lb/lb-mole

$$Md = (0.44 \times \%\ CO_2) + (0.32 \times \%\ O_2) + (0.28\ (100 - \%\ CO_2 - \%\ O_2))$$

where,

CO_2	0.0	= carbon dioxide concentration, %
O_2	20.8	= oxygen concentration, %
Md	28.83	= lb/lb mol

Molecular Weight (WET) (Ms), lb/lb-mole

$$Ms = Md\ (1 - BWS) + 18\ (BWS)$$

where,

Md	28.83	= molecular weight (DRY), lb/lb mol
BWS	0.007	= moisture fraction, dimensionless
Ms	28.75	= lb/lb mol

Average Velocity (Vs), ft/sec

$$Vs = 85.49 \times Cp \times (\Delta P^{1/2})\ avg \times \sqrt{\frac{Ts}{Ps\ x\ Ms}}$$

where,

Cp	0.840	= pitot tube coefficient
$\Delta P^{1/2}$	0.536	= velocity head of stack gas, (in. $H_2O)^{1/2}$
Ts	508.0	= absolute stack temperature, °R
Ps	28.13	= absolute stack gas pressure, in. Hg
Ms	28.75	= molecular weight of stack gas, lb/lb mol
Vs	30.5	= ft/sec

Average Stack Gas Flow at Stack Conditions (Qa), acfm

$$Qa = 60 \times Vs \times As$$

where,

Vs	30.5	= stack gas velocity, ft/sec
As	12.57	= cross-sectional area of stack, ft^2
Qa	23,007	= acfm

Average Stack Gas Flow at Standard Conditions (Qs), dscfm

$$Qs = 17.647 \times Qa \times (1 - BWS) \times \frac{Ps}{Ts}$$

where,

Qa	23,007	= average stack gas flow at stack conditions, acfm
BWS	0.007	= moisture fraction, dimensionless
Ps	28.13	= absolute stack gas pressure, in. Hg
Ts	508.0	= absolute stack temperature, °R
Qs	22,314	= dscfm

Dry Gas Meter Calibration Check (Yqa), dimensionless

$$Yqa = \frac{Y - \left(\frac{\Theta}{Vm} \sqrt{\frac{0.0319 \times Tm \times 29}{\Delta H@ \times \left(Pb + \frac{\Delta H\ avg.}{13.6} \right) \times Md}} \sqrt{\Delta H}\ avg. \right)}{Y} \times 100$$

where,

Y	0.986	= meter correction factor, dimensionless
Θ	60	= run time, min.
Vm	59.258	= total meter volume, dcf
Tm	533.7	= absolute meter temperature, °R
ΔH@	1.949	= orifice meter calibration coefficient, in. H_2O
Pb	28.74	= barometric pressure, in. Hg
ΔH avg	2.992	= average pressure differential of orifice, in H_2O
Md	28.83	= molecular weight (DRY), lb/lb mol
$(\Delta H)^{1/2}$	1.730	= average squareroot pressure differential of orifice, (in. $H2O)^{1/2}$
Yqa	2.2	= dimensionless

Dry Gas Meter Calibration Check (Yqa), dimensionless

$$Yqa = \frac{Y - \left(\frac{\Theta}{Vm} \sqrt{\frac{0.0319 \times Tm \times 29}{\Delta H @ \times \left(Pb + \frac{\Delta H\ avg.}{13.6} \right) \times Md}} \sqrt{\Delta H}\ avg. \right)}{Y} \times 100$$

where,

Y	0.99	= meter correction factor, dimensionless
Θ	120	= run time, min.
Vm	97.94426	= total meter volume, dcf
Tm	556.6	= absolute meter temperature, °R
ΔH@	2.003	= orifice meter calibration coefficient, in. H_2O
Pb	28.85	= barometric pressure, in. Hg
ΔH avg	2.208	= average pressure differential of orifice, in H_2O
Md	30.38	= molecular weight (DRY), lb/lb mol
$(\Delta H)^{1/2}$	1.486	= average squareroot pressure differential of orifice, $(\text{in. } H2O)^{1/2}$
Yqa	0.7	= dimensionless

Volume of Nozzle (Vn), ft^3

$$Vn = \frac{Ts}{Ps} \left(0.002669 \times Vlc + \frac{Vm \times Pm \times Y}{Tm} \right)$$

where,

Ts	589.5	= absolute stack temperature, °R
Ps	28.80	= absolute stack gas pressure, in. Hg
Vlc	386.3	= volume of H_2O collected, ml
Vm	97.944	= meter volume, cf
Pm	29.01	= absolute meter pressure, in. Hg
Y	0.990	= meter correction factor, unitless
Tm	556.6	= absolute meter temperature, °R
Vn	124.554	= volume of nozzle, ft^3

Isokinetic Sampling Rate (I), %

$$I = \left(\frac{Vn}{\theta \times 60 \times An \times Vs} \right) \times 100$$

where,

Vn	124.554	= nozzle volume, ft^3
θ	120.0	= run time, minutes
An	0.00028	= area of nozzle, ft^2
Vs	61.3	= average velocity, ft/sec
I	101.3	= %

Volume of Nozzle (Vn), ft^3

$$Vn = \frac{Ts}{Ps}\left(0.002669 \times Vlc + \frac{Vm \times Pm \times Y}{Tm}\right)$$

where,

Ts	508.0	= absolute stack temperature, °R
Ps	28.13	= absolute stack gas pressure, in. Hg
Vlc	8.8	= volume of H_2O collected, ml
Vm	59.258	= meter volume, cf
Pm	28.96	= absolute meter pressure, in. Hg
Y	0.986	= meter correction factor, unitless
Tm	533.7	= absolute meter temperature, °R
Vn	57.690	= volume of nozzle, ft^3

Isokinetic Sampling Rate (I), %

$$I = \left(\frac{Vn}{\theta \times 60 \times An \times Vs}\right) \times 100$$

where,

Vn	57.690	= nozzle volume, ft^3
θ	60.0	= run time, minutes
An	0.00052	= area of nozzle, ft^2
Vs	30.5	= average velocity, ft/sec
I	101.5	= %

Filterable PM Concentration (C_s), grain/dscf

$$C_s = \frac{M_n \times 0.0154}{Vmstd}$$

where,

Mn	7.0	= filterable PM mass, mg
Vmstd	55.953	= standard meter volume, dscf
C_s	0.0019	= grain/dscf

REFERENCE METHOD DATA

TEDLAR BAG

Molecular Weight

Method 3-3A Tedlar Bag

Analyzer Calibration

Calibration Date/Time			Analyzer Make/Model/SN		
2/1/2025	8:00		Teledyne	T802	156

		O_2 Data		CO_2 Data	
	Cylinder ID	Cylinder Concentration (%)	Analyzer Response (%)	Cylinder Concentration (%)	Analyzer Response (%)
Zero Gas	UHP Nitrogen	0.00	0.00	0.00	0.00
Span Cal Gas	DT0031874	20.00	20.01	20.10	20.09
Mid Cal Gas	CC199274	9.98	10.01	9.61	9.70
	Required Accuracy	± 0.4% of cylinder concentration		± 0.4% of cylinder concentration	

Tedlar Bag Analysis

Tedlar Bag ID	Start	End	% O_2	% CO_2
Run 1	10:19	13:05	20.81	0.00

Tedlar Bag ID	Start	End	% O_2	% CO_2
Run 2	14:04	15:09	20.79	0.00

Tedlar Bag ID	Start	End	% O_2	% CO_2
Run 3	15:20	16:24	20.78	0.00

PARTICULATE MATTER FIELD DATA

Evergy - Field Data

Plant, Unit:	#4 Dust Collector	**Barometric Pressure:**	28.74 inHG	**Leak Check, I:**	0.000 @ 15 inHG
Run #:	1	**Static Pressure:**	-8.34 inH2O	**Leak Check, F:**	0.000 @ 7 inHG
Date:	2/1/2025	**Stack Diameter/Area:**	48 in. / 12.57 sqft		
Operator:	BJ	**itot Tube Coefficient:**	0.84	**Pitot Leak Check, I:**	Pass
Test Type:	M5	**DGM Coefficient (Y):**	0.986	**Pitot Leak Check, F:**	Pass
Run Time:	60 minutes	**Delta H @:**	1.949		
Start/End:	10:19 / 13:05	**Meter Box #:**	1806015	**Probe ID:**	Glass M5 11ft A
	Estimated / Final	**Nozzle Diameter:**	0.308	**Pitot ID:**	A8481
O_2, %:	20.90 / 20.81	**Assumed % Moisture:**	2	**Filter ID Number:**	5B429
CO_2, %:	0.00 / 0.00	**Final % Moisture:**	0.7 Actual	**Project Number:**	25-003

Point	Sample Time	DGM (ft^3)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F		Probe Temp °F		Filter Temp. °F		CPM Filter Temp. °F		Cold Box Temp.	DGM Temp. °F
A-1	10:19	413.396	3.10	7	0.30	47		250		250		NA		37	65
2	10:24	418.340	3.10	7	0.29	47		256		249		NA		37	67
3	10:29	423.280	2.90	7	0.28	47		252		251		NA		39	69
B-1	10:34	428.370	2.90	7	0.28	47		244		249		NA		41	71
2	10:39	433.230	2.80	7	0.27	47		248		250		NA		42	72
3	10:44	437.970	3.00	7	0.29	47		251		250		NA		42	75
C-1	10:55	442.905	3.10	7	0.30	47		250		251		NA		43	76
2	11:00	448.170	3.10	7	0.29	47		249		250		NA		43	78
3	11:05	452.955	2.90	7	0.28	47		250		249		NA		43	78
D-1	12:50	457.750	3.10	7	0.30	49		249		251		NA		41	77
2	12:55	462.765	3.00	7	0.29	49		245		252		NA		41	78
3	13:00	467.720	2.90	7	0.28	49		246		247		NA		43	78
End	13:05	472.654													
		Total ft^3:	Avg:	High:	Avg Sq. Rt.:	High:	Low:	High:	Low:	High:	Low:	High:	Low:	High:	Avg:
		59.258	2.99	7	0.5361	49	47	256	244	252	247			43	74

Initial Wt: 2960.6

Final Wt: 2969.4

Weight Gain: 8.8

DGM Calibration Check (Yqa): 2.2

(Yqa must be ± 5 to be valid)

K-Factor: 10.35

% ISO: 101.5

Evergy - Field Data

Plant, Unit:	#4 Dust Collector	**Barometric Pressure:**	28.74 inHG	**Leak Check, I:**	0.000 @ 15 inHG
Run #:	2	**Static Pressure:**	-8.34 inH2O	**Leak Check, F:**	0.000 @ 7 inHG
Date:	2/1/2025	**Stack Diameter/Area:**	48 in. / 12.57 sqft		
Operator:	BJ	**itot Tube Coefficient:**	0.84	**Pitot Leak Check, I:**	Pass
Test Type:	M5	**DGM Coefficient (Y):**	0.986	**Pitot Leak Check, F:**	Pass
Run Time:	60 minutes	**Delta H @:**	1.949		
Start/End:	14:04 / 15:09	**Meter Box #:**	1806015	**Probe ID:**	Glass M5 11ft B
	Estimated / Final	**Nozzle Diameter:**	0.303	**Pitot ID:**	A8482
O_2, %:	20.90 / 20.79	**Assumed % Moisture:**	2	**Filter ID Number:**	5B430
CO_2, %:	0.00 / 0.00	**Final % Moisture:**	0.8 Actual	**Project Number:**	25-003

Point	Sample Time	DGM (ft^3)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cold Box Temp	DGM Temp. °F
A-1	14:04	475.920	2.80	7	0.28	48	254	253	NA	35	77
2	14:09	480.500	2.70	7	0.27	49	248	249	NA	37	77
3	14:14	485.160	2.70	7	0.27	50	252	252	NA	41	79
B-1	14:19	489.910	2.80	7	0.29	50	250	248	NA	42	79
2	14:24	494.825	2.90	7	0.29	50	250	251	NA	43	81
3	14:29	499.555	2.80	7	0.28	50	252	252	NA	44	82
C-1	14:39	504.345	2.70	7	0.27	50	247	250	NA	45	84
2	14:44	509.110	2.80	7	0.28	50	249	250	NA	46	85
3	14:49	513.905	2.70	7	0.27	50	248	250	NA	46	86
D-1	14:54	518.705	3.00	7	0.30	50	248	249	NA	47	87
2	14:59	523.640	2.90	7	0.29	50	251	251	NA	48	88
3	15:04	528.525	2.80	7	0.28	51	249	249	NA	49	88
End	15:09	533.337									
		Total ft^3:	Avg:	High:	Avg Sq. Rt.:	High: / Low:	High: / Low:	High: / Low:	High: / Low:	High:	Avg:
		57.417	2.80	7	0.5299	51 / 48	254 / 247	253 / 248	/	49	83

Initial Wt: 2969.4
Final Wt: 2978
Weight Gain: 8.6

DGM Calibration Check (Yqa): 1.5
(Yqa must be ± 5 to be valid)

K-Factor: 9.82
% ISO: 101.2

Evergy - Field Data

Plant, Unit:	#4 Dust Collector	**Barometric Pressure:**	28.74 inHG	**Leak Check, I:**	0.000 @ 15 inHG
Run #:	3	**Static Pressure:**	-8.34 inH2O	**Leak Check, F:**	0.000 @ 7 inHG
Date:	2/1/2025	**Stack Diameter/Area:**	48 in. / 12.57 sqft		
Operator:	BJ	**itot Tube Coefficient:**	0.84	**Pitot Leak Check, I:**	Pass
Test Type:	M5	**DGM Coefficient (Y):**	0.986	**Pitot Leak Check, F:**	Pass
Run Time:	60 minutes	**Delta H @:**	1.949		
Start/End:	15:20 / 16:24	**Meter Box #:**	1806015	**Probe ID:**	Glass M5 11ft A
	Estimated / **Final**	**Nozzle Diameter:**	0.308	**Pitot ID:**	A8481
O_2, %:	20.90 / 20.77	**Assumed % Moisture:**	2	**Filter ID Number:**	5B431
CO_2, %:	0.00 / 0.00	**Final % Moisture:**	0.8 Actual	**Project Number:**	25-003

Point	Sample Time	DGM (ft^3)	Delta H	Vacuum (in. Hg)	Delta P	Stack Temp. °F	Probe Temp °F	Filter Temp. °F	CPM Filter Temp. °F	Cold Box Temp	DGM Temp. °F
A-1	15:20	535.132	2.80	7	0.29	50	249	250	NA	39	87
2	15:25	540.190	3.00	7	0.28	50	251	247	NA	40	88
3	15:30	545.315	3.00	7	0.28	50	252	255	NA	42	88
B-1	15:35	550.370	3.10	7	0.29	50	252	249	NA	43	88
2	15:40	555.410	2.90	7	0.27	50	251	247	NA	44	89
3	15:45	560.215	3.00	7	0.28	50	248	250	NA	45	90
C-1	15:54	565.330	3.10	7	0.29	50	250	250	NA	45	90
2	15:59	570.280	3.20	7	0.30	50	249	251	NA	46	90
3	16:04	575.505	2.90	7	0.27	50	249	250	NA	47	91
D-1	16:09	580.365	3.10	7	0.29	50	248	249	NA	49	91
2	16:14	585.590	3.10	7	0.29	50	247	251	NA	51	91
3	16:19	590.640	3.00	7	0.28	50	252	249	NA	52	92
End	16:24	595.890									
		Total ft^3: 60.758	Avg: 3.02	High: 7	Avg Sq. Rt.: 0.5330	High: 50 / Low: 50	High: 252 / Low: 247	High: 255 / Low: 247	High: / Low:	High: 52	Avg: 90

Initial Wt: 2968.6
Final Wt: 2978.4
Weight Gain: 9.8

DGM Calibration Check (Yqa): 2.8
(Yqa must be ± 5 to be valid)

K-Factor: 10.64
% ISO: 101.9

Plant/Unit: #4 Dust Collector
Project No.: 25-003

Method 4 Moisture Determination

Impinger Weights

		Contents	Initial Weight	Final Weight	Difference
Run 1	#1	DI Water	710	709.3	-0.7
	#2	DI Water	681	685.8	4.8
	#3	Empty	602	604.9	2.9
	#4	Silica	967.6	969.4	1.8
		Total:	2960.6	2969.4	
Run 2	#1	DI Water	709.3	710.2	0.9
	#2	DI Water	685.8	689.8	4
	#3	Empty	604.9	607.7	2.8
	#4	Silica	969.4	970.3	0.9
		Total:	2969.4	2978	
Run 3	#1	DI Water	707.5	708.9	1.4
	#2	DI Water	685.7	690.3	4.6
	#3	Empty	605.1	607.6	2.5
	#4	Silica	970.3	971.6	1.3
		Total:	2968.6	2978.4	

Evergy - Field Data

Plant/Unit: #4 Dust Collector Project Number: 25-003
Operator: BJ Date: 2/1/2025

Nozzle Set	Nozzle ID	D_n (in)		
G-2	10	0.308		
G-1	10	0.303		
Pitot ID	**Evidence of damage?**	**Evidence of mis-alignment?**	**Calibration or Repair required?**	
A8481	No	No	No	
A8482	No	No	No	
Probe ID	**Reference Temp. (°F)**	**Indicated Temp. (°F)**	**Difference**	**Criteria**
Glass M5 11ft A	47.0	48.0	0.2%	± 1.5 % (absolute)
Glass M5 11ft B	47.0	47.0	0.0%	
Barometric Reading from:	**Barometric Reading @ MSL (inHG)**	**Barometric Reading @ Elevation**	**Elevation (ft)**	
NOAA	28.74	28.74	15	

INSTRUMENT VALIDATION CHECK

TELEDYNE
ADVANCED POLLUTION INSTRUMENTATION
A Teledyne Technologies Company

T802 Final Calibrated Test and Validation Data

5 12/11/18

Model:	T802		
Firmware:	063020001 1.0.2.73	Serial Number: 155	Sales Order:
Date:	12/10/2018	Technician: FG	SP#: YES

Parameter	Displayed As	Observed Value	Units	Final Test Process Control Limits at Factory**
O_2 Range	O_2 RNG	0-25	WT%	0 - 1 WT% to 1 - 100 WT%
CO_2 Range (Opt only)	CO_2 RNG	0-20	WT%	0 - 20 WT% Standard
Stability	O_2 STB	0	WT%	< 0.05 % with Zero Air
Sample Pressure	PRES	29.2	In-Hg-A	Ambient Pressure ± 1.0 In Hg
Sample Flow	SAMP FL	138	cc/min	120 ± 20
O_2 Slope	O_2 SLOPE	1.009	-	1.0 ± 0.1
O_2 Offset	O_2 OFST	0.27	%	≤ 1%
CO_2 Slope (Opt Only)	CO_2 SLOPE	1.015	-	1.0 ± 0.1
CO_2 Offset (Opt Only)	CO_2 OFST	0.08	%	≤ 1%
O_2 Cell Temperature *	O_2 CELL TEMP	50	°C	50 ± .5
CO_2 Cell Temp * (Opt Only)	CO_2 CELL TEMP	50	°C	50 ± .5
Box Temperature	BOX TEMP	32.9	°C	31 ± 4
Time of Day	TIME	13:59:57	HH:MM:SS	

* For good instrument performance, the steadiness of this signal is more important than its absolute value (within the operating range)
** these are process control limits, and not specification limits. Items out of range do not imply the unit is out of specification.

Statement of Calibration
The unit identified above has been tested with NIST measuring and test equipment using lot traceable materials. The testing is performed in accordance with ISO 9001-2015 and is traceable to NIST and industry recognized standards.

Configuration and Options		
Power Configuration: Voltage	100-125	
Frequency	60 Hz	
HV internal pump (13)		
Rack Mounts with Slides (20A - 26"; 20B - 24")		
Rack Mounts Ears Only (21)		
Strap Carrying Handle (29)		
Multi-drop RS-232 (62)		
Ethernet (63A)	Yes	
Multidrop/ethernet combo (63C)		
USB Com 2 T-series only (64A)		
Analog Input T-series only (64B)		
COM 2 Configuration		
Warranty Extended 1 Year (92A)		
A1 Voltage Output		
A2 Voltage Output		
A3 Voltage Output		
A4 Voltage Output	5	
Current Output, Channel A1 (41)	4 - 20 mA	
Current Output, Channel A2 (41)	4 - 20 mA	
Current Output, Channel A3 (41)		

Additional Information:

TML, OPT, SLIDES WITH EXT, PN: 023600100. NUMAVIEW FW, PN: 082910000 1.4.4.115. OPT 67A, CO2 SENSOR PN: 064790000

06835 Rev B (DCN 5976) 1/25/2011 Page 1 of 1

CALIBRATION GAS CERTIFICATIONS

PRAXAIR

Making our planet more productive

DocNumber: 413305

Praxair Distribution, Inc.
6055 Brent Drive
Toledo OH 43611
Tel: +1 (419) 729-7732
Fax: +1 (419) 729-2411
PGVP ID: F12021

CERTIFICATE OF ANALYSIS / EPA PROTOCOL GAS

Customer & Order Information

PRAXAIR PKG LENEXA KS P
9725 ALDEN RD
LENEXA KS 66215-1128

Certificate Issuance Date: 07/02/2021
Praxair Order Number: 71733991
Part Number: NI CD20O15E-AS
Customer PO Number: 79732586

Fill Date: 06/16/2021
Lot Number: 70001116?F4
Cylinder Style & Outlet: AS CGA 590
Cylinder Pressure and Volume: 2000 psig 140 ft3

Certified Concentration

Expiration Date:	**07/01/2029**	NIST Traceable
Cylinder Number:	**DT0031874**	Expanded Uncertainty
20.1 %	**Carbon dioxide**	**± 0.1 %**
20.00 %	**Oxygen**	**± 0.01 %**
Balance	**Nitrogen**	

ProSpec EZ Cert

Certification Information:

Certification Date: 07/01/2021 *Term:* 96 Months *Expiration Date:* 07/01/2029

This cylinder was certified according to the 2012 EPA Traceability Protocol, Document #EPA-600/R-12/531, using Procedure G1. Uncertainty above is expressed as absolute expanded uncertainty at a level of confidence of approximately 95% with a coverage factor k = 2. Do Not Use this Standard if Pressure is less than 100 PSIG.

O2 responses have been corrected for CO2 interference.

Analytical Data:

(R=Reference Standard, Z=Zero Gas, C=Gas Candidate)

1. Component: **Carbon dioxide**

Requested Concentration: 20.0 %
Certified Concentration: 20.1 %
Instrument Used: MKS 2031
Analytical Method: FTIR
Last Multipoint Calibration: 06/07/2021

Reference Standard: Type / Cylinder #: GMIS / DT0033247
Concentration / Uncertainty: 19.99 % ±0.04 %
Expiration Date: 01/02/2028
Traceable to: SRM # / Sample # / Cylinder #: PRM / 3222577.01 / FF27013
SRM Concentration (enter with units) / Uncertainty: 20.008% / ±0.028
SRM Expiration Date: 04/01/2020

First Analysis Data:						**Date**	07/01/2021
Z:	0	**R:**	20.1	**C:**	20.2	**Conc:**	20.2
R:	20	**Z:**	0	**C:**	20.1	**Conc:**	20.1
Z:	0	**C:**	20.1	**R:**	20	**Conc:**	20.1
UOM:	ppm			**Mean Test Assay:**		20.1	%

Second Analysis Data:						**Date**	
Z:	0	**R:**	0	**C:**	0	**Conc:**	0
R:	0	**Z:**	0	**C:**	0	**Conc:**	0
Z:	0	**C:**	0	**R:**	0	**Conc:**	0
UOM:	ppm			**Mean Test Assay:**			%

2. Component: **Oxygen**

Requested Concentration: 20.0 %
Certified Concentration: 20.00 %
Instrument Used: Servomex 575
Analytical Method: Paramagnetic
Last Multipoint Calibration: 06/03/2021

Reference Standard: Type / Cylinder #: GMIS / CC257802
Concentration / Uncertainty: 22.5 % ±0.0 %
Expiration Date: 12/02/2027
Traceable to: SRM # / Sample # / Cylinder #: 2659a / 71-D-04 / CAL015785
SRM Concentration (enter with units) / Uncertainty: 20.72% / ±0.043%
SRM Expiration Date: 08/23/2021

First Analysis Data:						**Date**	07/01/2021
Z:	0	**R:**	22.5	**C:**	20	**Conc:**	20
R:	22.5	**Z:**	0	**C:**	20	**Conc:**	20
Z:	0	**C:**	20	**R:**	22.5	**Conc:**	20
UOM:	%			**Mean Test Assay:**		20	%

Second Analysis Data:						**Date**	
Z:	0	**R:**	0	**C:**	0	**Conc:**	0
R:	0	**Z:**	0	**C:**	0	**Conc:**	0
Z:	0	**C:**	0	**R:**	0	**Conc:**	0
UOM:	%			**Mean Test Assay:**			%

Analyzed By: Mike Mannette

Certified By: Edward E Zetal

Making our world more productive

DocNumber: 532768

Linde Gas & Equipment Inc.
6055 Brent Drive
Toledo OH 43611
Tel: +1 (419) 729-7732
Fax: +1 (419) 729-2411
PGVP ID: F12023

CERTIFICATE OF ANALYSIS / EPA PROTOCOL GAS

Customer & Order Information

LGEPKG LENEXA KS P
9725 ALDEN RD
LENEXA KS 66215-1128

Certificate Issuance Date: 01/26/2023
Linde Order Number: 72317322
Part Number: NI CD10O10E-AS
Customer PO Number: 80324183

Fill Date: 01/10/2023
Lot Number: 700013010F7
Cylinder Style & Outlet: AS CGA 590
Cylinder Pressure and Volume: 2000 psig 140 ft3

Certified Concentration

Expiration Date:	01/26/2031	NIST Traceable
Cylinder Number:	CC199274	Expanded Uncertainty
9.61 %	Carbon dioxide	± 0.10 %
9.98 %	Oxygen	± 0.04 %
Balance	Nitrogen	

ProSpec EZ Cert

Certification Information:

Certification Date: 01/26/2023 *Term:* 96 Months *Expiration Date:* 01/26/2031

This cylinder was certified according to the 2012 EPA Traceability Protocol, Document #EPA-600/R-12/531, using Procedure G1. Uncertainty above is expressed as absolute expanded uncertainty at a level of confidence of approximately 95% with a coverage factor k = 2. Do Not Use this Standard if Pressure is less than 100 PSIG.

O2 responses have been corrected for CO2 interference.

Analytical Data:

(R=Reference Standard, Z=Zero Gas, C=Gas Candidate)

1. Component: Carbon dioxide

Requested Concentration: 10.0 %
Certified Concentration: 9.61 %
Instrument Used: Horiba VA-5000
Analytical Method: NDIR
Last Multipoint Calibration: 12/28/2022

Reference Standard: Type / Cylinder #: GMIS / DT0032770
Concentration / Uncertainty: 20.0 % ±0.0 %
Expiration Date: 01/02/2028
Traceable to: SRM # / Sample # / Cylinder #: PRM# FF276163 / PRM# FF276163 / FF276163
SRM Concentration / Uncertainty: 20.008% / ±0.028%
SRM Expiration Date: 04/01/2020

First Analysis Data:						Date	01/26/2023
Z:	0	R:	20	C:	9.6	Conc:	9.6
R:	20	Z:	0	C:	9.61	Conc:	9.61
Z:	0	C:	9.61	R:	20.01	Conc:	9.61
UOM:	%				Mean Test Assay:	9.61	%

Second Analysis Data:						Date	
Z:	0	R:	0	C:	0	Conc:	0
R:	0	Z:	0	C:	0	Conc:	0
Z:	0	C:	0	R:	0	Conc:	0
UOM:	%				Mean Test Assay:		%

2. Component: Oxygen

Requested Concentration: 10.0 %
Certified Concentration: 9.98 %
Instrument Used: Servomex 575
Analytical Method: Paramagnetic
Last Multipoint Calibration: 01/06/2023

Reference Standard: Type / Cylinder #: SRM / DT0032356
Concentration / Uncertainty: 22.52 % ±0.01 %
Expiration Date: 12/02/2027
Traceable to: SRM # / Sample # / Cylinder #: 2659a / 71-D-04 / CAL015785
SRM Concentration / Uncertainty: 20.720% / ±.043%
SRM Expiration Date: 08/23/2021

First Analysis Data:						Date	01/26/2023
Z:	0	R:	22.52	C:	9.98	Conc:	9.98
R:	22.51	Z:	0	C:	9.98	Conc:	9.98
Z:	0	C:	9.99	R:	22.52	Conc:	9.99
UOM:	%				Mean Test Assay:	9.98	%

Second Analysis Data:						Date	
Z:	0	R:	0	C:	0	Conc:	0
R:	0	Z:	0	C:	0	Conc:	0
Z:	0	C:	0	R:	0	Conc:	0
UOM:	%				Mean Test Assay:		%

Analyzed By Lloyd Knapp

Certified By Edward E Zucal

Issue Date: **6/27/2019**

PRAXAIR

To: Evergy

Attn: Alan Eason

CERTIFICATE OF CONFORMANCE

Business Confidential

Dear Customer:

This is to advise that UHP Grade Nitrogen, (NI 5.0) supplied to Evergy, Topeka, KS by Praxair Distribution, Lenexa, KS meets or exceeds the following minimum purity and maximum impurity specification content:

Cylinder Serial Numbers: UHP Nitrogen
Nitrogen: 99.999% minimum purity
O_2: 5 ppm maximum
THC: 0.5 ppm maximum
H_2O: 3 ppm maximum

Sincerely,

Todd Hallier
Praxair Distribution Inc.
1608 Holmes
Kansas City MO 64108

EQUIPMENT CALIBRATION RECORDS

Dry Gas Meter Calibration

Console ID:	1806015
Date:	01/02/24

Current Cal:

Y_d = 0.983	
$\Delta H_@$ = 1.930	

Previous Cal:

Y_d =	0.989
$\Delta H_@$ =	1.936

Pass: X

Fail:

Orifice ID:	UY-48	UY-55	UY-63
Orifice K':	0.3451	0.4476	0.5822
	Run 1	Run 2	Run 3
Initial Reading, (ft^3):	792.300	785.800	777.500
Final Reading, (ft^3):	799.219	791.780	785.255
Difference, (ft^3) [V_w]:	6.919	5.980	7.755
Initial Meter Temp., (°F):	75	74	73
Final Meter Temp., (°F):	75	74	74
Average Meter Temp., (°F):	75.0	74.0	73.5
Test Time (min.) [q]:	15.00	10.00	10.00
Orifice Manometer Reading, (in H_2O):	0.69	1.15	1.90
Barometric Pressure, (in HG):	29.30	29.30	29.31
Ambient Temperature, (°F):	69.9	69.5	68.5
Pump Vacuum, (in Hg):	24.5	23.5	22
Standard Volume of the Meter, $V_{m(std)}$ =	6.696	5.805	7.551
Standard Volume of Critical Orifice, $V_{cr(std)}$ =	6.589	5.699	7.423
DGM Calibration Factor, (Y) =	0.984	0.982	0.983
$\Delta H_@$ =	1.951	1.939	1.899
Percent Difference =	-0.109	0.111	-0.002

Average Y =	0.983	1.000 ± .03 acceptability criteria
Reference Yd =	1.000	
Average $\Delta H_@$ =	1.930	
STD =	0.00108	
Percent Difference (Previous Cal)	-0.62	± 5% acceptability criteria

Operator: Samuel Burt

Date: 1/2/2024

Calibration Pyrometer Check

	Cal Temp °F	Pyrometer Temp °F	Percent difference
	0	2	0.4
Ambient temp	68	68	0.0
	200	202	0.3
	250	252	0.3
	300	302	0.3
	350	351	0.1
	400	400	0.0
	500	500	0.0
	Max Devation (%)		0.4

V&A Instruments Model VA710 Calibrator-Thermometer SN: VA181203023 evergy

Nozzle Calibration

Set ID: G-1

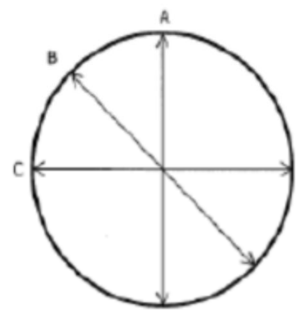
Nozzle Type: Glass

Nozzle ID:	4	5	6	7	8	9	10
Diameter A:	0.122	0.157	0.186	0.226	0.256	0.287	0.304
Diameter B:	0.123	0.157	0.186	0.225	0.256	0.286	0.302
Diameter C:	0.122	0.157	0.185	0.226	0.256	0.289	0.303
Effective Diameter:	0.122	0.157	0.186	0.226	0.256	0.287	0.303

Acceptability Criteria: 40 CFR 60 Method 5.10.1 - The difference between the high and low readings shall not exceed 0.004 in. (0.1 mm).

Calibrated by: 1/3/25

(Signature/Date)



evergy

Nozzle Calibration

Set ID: G-2

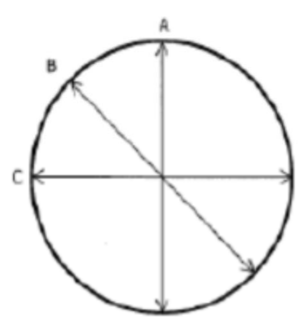
Nozzle Type: Glass

Nozzle ID:	4	5	6	7	8	9	10
Diameter A:	0.126	0.155	0.190	0.222	0.257	0.284	0.308
Diameter B:	0.126	0.154	0.190	0.222	0.257	0.285	0.307
Diameter C:	0.125	0.155	0.188	0.220	0.256	0.286	0.308
Effective Diameter:	0.126	0.155	0.189	0.221	0.257	0.285	0.308

Acceptability Criteria: 40 CFR 60 Method 5.10.1 - The difference between the high and low readings shall not exceed 0.004 in. (0.1 mm).

Calibrated by: 1/3/2025

(Signature/Date)



Public

ET&A ACCREDITATION AND QSTI CERTIFICATIONS

Date: April 1, 2019

To: Environmental Group, Evergy

From: Dan Wilkus, Senior Manager of Environmental Services

Subject: Certification of Compliance with ASTM D 7036-04

As of April 1, 2019, Evergy Testing & Analytics (ET&A) group has self-accredited as an Air Emissions Testing Body (AETB). This letter is to certify that our testing methods and implemented processes will meet the requirements of Federal, State, and local regulations. As required by ASTM D 7036-04 (Standard Practice for Competence of Air Emissions Testing Bodies) and 40 CFR 75 (Continuous Emissions Monitoring) we have developed and emissions testing procedure and will abide by all requirements noted therein.

Dan Wilkus
Senior Manager of Environmental Services

Jeremy Duis
Manager CEMs and Reporting

SOURCE EVALUATION SOCIETY

Qualified Source Testing Individual

LET IT BE KNOWN THAT

BRANDON M. JONES

HAS SUCCESSFULLY PASSED A COMPREHENSIVE EXAMINATION AND SATISFIED EXPERIENCE REQUIREMENTS IN ACCORDANCE WITH THE GUIDELINES ISSUED BY THE SES QUALIFIED SOURCE TEST INDIVIDUAL REVIEW BOARD FOR

PART 75 CEMS RATA TESTING

ISSUED THIS 9TH DAY OF APRIL 2024 AND EFFECTIVE UNTIL APRIL 8TH, 2029

Peter R. Westlin, QSTI/QSTO Review Board

J. Wade Bice, QSTI/QSTO Review Board

Peter S. Pakalnis, QSTI/QSTO Review Board

Karen D. Kajiya-Mills , QSTI/QSTO Review Board

Tina Sanderson, QSTI/QSTO Review Board

Bruce Randall QSTI/QSTO Review Board

CERTIFICATE
NO.
2024-1152

PARTICULATE MATTER
LAB REPORT

JEFFERY ENERGY CENTER – DUST BIN COLLECTOR #4

ET&A PROJECT 25-003
DATE: 2/28/2025

I certify that to the best of my knowledge all analytical data presented in this report:

- Have been checked for completeness
- Are accurate, error-free, and legible
- Have been conducted in accordance with approved protocol, and that all deviations and analytical problems are summarized in the appropriate narrative

2/28/2025

Signature / Date

ANALYTICAL NARRATIVE SUMMARY

Chain of Custody

Relinquished By: ON SITE
Date and Time:

Received By: ON SITE
Date and Time:

Analysis

- The samples were analyzed for particulate matter and condensable particulate matter using the analytical procedures in USEPA Method 5. All samples were analyzed following the procedures in section 11.0, Analytical Procedure.

Calibration

- The analytical balance was calibrated using certified class "S" weights prior to analysis of any samples. For each calibration point, the balance was required to read within 0.5 mg of the actual certified mass of the class "S" weight. The calibration range bracketed the lower and upper mass of the samples. A copy of the calibration log is included in this report.

The analytical environment met the following conditions:

- Temperature: 58-78 °F
- Humidity: < 55%

QC Notes

- An Acetone Blank was prepared and analyzed with the samples. The blank value is included in the Beaker Tare Log of this report.

No analytical discrepancies occurred.

All Samples were processed on a A&D HR-100AZ Analytical Balance with a weighing capacity of 102 grams and readability to 0.0001 grams. The balance is certified annually.

SUMMARY OF RESULTS

Method 5 Laboratory Data

Beaker Tare Weight (g):	51.7665	51.8116	53.2783	
Beaker Final Weight (g):	51.7732	51.8184	53.2877	
Filterable PM mass (probe) (g):	0.0067	0.0068	0.0094	0.00763
Filter Tare Weight (g):	0.3524	0.3528	0.3525	
Filter Final Weight (g):	0.3527	0.3531	0.3532	
Filterable PM mass (filter) (g):	0.0003	0.0003	0.0007	0.00043
Acetone Wash (ml):	56	59	60	

Blank Volume (ml)	200
Reagent Blank (g)	0.00010
Acetone Density g/l	0.7846
Max Ca%residue	0.0010%
Ca%residue	0.0001%

CALIBRATION LOG

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Troemner, LLC

ISO/IEC 17025 Calibration Certificate

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Weight

ID Number	4000022685
Certificate Number	01061080-1
Date of Calibration	25-FEB-2019

SECTION 1: NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 2: APPROVED SIGNATORY

Annemarie Love, Metrologist

SECTION 3: PERSON PERFORMING WORK

Robotic Calibration

SECTION 4: CERTIFICATE INFORMATION

Description of Masses: ASTM Weight Set

Accuracy Class	: ASTM E617-13 Class 1	Date Received	: 19-FEB-2019
Order Number	: 0074451-00	Date of Calibration	: 25-FEB-2019
Construction	: One Piece, Two Piece	Date of Issue	: 25-FEB-2019
Material	: Aluminum	Weight Range	: 1mg-2mg
	: Stainless Steel		: 5mg-20g
Serial Number	: 4000022685		

SECTION 5: ENVIRONMENTAL CONDITIONS DURING TEST

Temperature: 21.46 °C Pressure: 757.60 mm Hg Relative Humidity: 48%

SECTION 6: PERTINENT INFORMATION

The Weights listed on this calibration report have been compared to reference mass standards that are traceable to the SI through the National Institute of Standards and Technology under Test No. 684/289871-17.

Reference standards and balances used to perform the calibration are listed in Section 10.

The weights calibrated for this report have been calibrated in accordance with Troemner's calibration process. The calibration performed meets the criteria as described in the current revisions of ASTM E617 and OIML R111.

This calibration also meets specifications as outlined in ISO/IEC 17025, ANSI/NCSL Z540-1-1994, and applicable documents.

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NVLAP CALIBRATION NVLAP LAB CODE 105013-0

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Weight

ID Number	4000022685
Certificate Number	01061080-1
Date of Calibration	25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 7: TRUE MASS (MASS IN VACUUM) CALIBRATION DATA

Nominal Mass Value	Serial Number	True Mass	Density [1] of Weight	Uncertainty (+ or -)
20 g		19.999980 g	8.0300 g/cm³	0.015 mg
10 g		9.999985 g	8.0300 g/cm³	0.010 mg
5 g		5.0000007 g	8.0300 g/cm³	0.0070 mg
2 g		2.0000008 g	8.0300 g/cm³	0.0060 mg
2 g *		1.9999994 g	8.0300 g/cm³	0.0060 mg
1 g		0.9999994 g	8.0300 g/cm³	0.0060 mg
500 mg		0.5000046 g	7.9500 g/cm³	0.0025 mg
200 mg		0.2000057 g	7.9500 g/cm³	0.0025 mg
200 mg *		0.1999996 g	7.9500 g/cm³	0.0025 mg
100 mg		0.1000005 g	7.9500 g/cm³	0.0025 mg
50 mg		0.0500061 g	7.9500 g/cm³	0.0025 mg
20 mg		0.0200033 g	7.9500 g/cm³	0.0020 mg
20 mg *		0.0199995 g	7.9500 g/cm³	0.0020 mg
10 mg		0.0100039 g	7.9500 g/cm³	0.0020 mg
5 mg		0.0050036 g	7.9500 g/cm³	0.0020 mg
2 mg		0.0020033 g	2.7000 g/cm³	0.0020 mg
2 mg *		0.0020025 g	2.7000 g/cm³	0.0020 mg
1 mg		0.0010037 g	2.7000 g/cm³	0.0020 mg

1 Density is assumed unless otherwise stated
* Denotes weight is marked with a dot.

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Weight
ID Number 4000022685
Certificate Number 01061080-1
Date of Calibration 25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 8: CONVENTIONAL MASS CALIBRATION VALUE VS. REFERENCE DENSITY 8000 kg/m³

Nominal Mass Value	Serial Number	Conventional Mass Value	Uncertainty (+ or -)	Tolerance (+ or -)
20 g		19.999992 g	0.015 mg	0.0740 mg
10 g		9.999991 g	0.010 mg	0.0500 mg
5 g		5.0000035 g	0.0070 mg	0.0340 mg
2 g		2.0000019 g	0.0060 mg	0.0340 mg
2 g *		2.0000005 g	0.0060 mg	0.0340 mg
1 g		0.9999999 g	0.0060 mg	0.0340 mg
500 mg		0.5000041 g	0.0025 mg	0.0100 mg
200 mg		0.2000055 g	0.0025 mg	0.0100 mg
200 mg *		0.1999994 g	0.0025 mg	0.0100 mg
100 mg		0.1000004 g	0.0025 mg	0.0100 mg
50 mg		0.0500060 g	0.0025 mg	0.0100 mg
20 mg		0.0200033 g	0.0020 mg	0.0100 mg
20 mg *		0.0199995 g	0.0020 mg	0.0100 mg
10 mg		0.0100039 g	0.0020 mg	0.0100 mg
5 mg		0.0050036 g	0.0020 mg	0.0100 mg
2 mg		0.0020027 g	0.0020 mg	0.0100 mg
2 mg *		0.0020019 g	0.0020 mg	0.0100 mg
1 mg		0.0010034 g	0.0020 mg	0.0100 mg

* Denotes weight is marked with a dot.

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NVLAP CALIBRATION NVLAP LAB CODE 105013-0 ilac-MRA

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Weight
ID Number 4000022685
Certificate Number 01061080-1
Date of Calibration 25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 9: CONVENTIONAL MASS CALIBRATION DATA VS. REFERENCE DENSITY 8000 kg/m³

Nominal Mass Value	Serial Number	Conventional Mass Correction	Uncertainty (+ or -)	Tolerance (+ or -)
20 g		-0.009 mg	0.015 mg	0.0740 mg
10 g		-0.009 mg	0.010 mg	0.0500 mg
5 g		0.0035 mg	0.0070 mg	0.0340 mg
2 g		0.0019 mg	0.0060 mg	0.0340 mg
2 g *		0.0005 mg	0.0060 mg	0.0340 mg
1 g		-0.0001 mg	0.0060 mg	0.0340 mg
500 mg		0.0041 mg	0.0025 mg	0.0100 mg
200 mg		0.0055 mg	0.0025 mg	0.0100 mg
200 mg *		-0.0006 mg	0.0025 mg	0.0100 mg
100 mg		0.0004 mg	0.0025 mg	0.0100 mg
50 mg		0.0060 mg	0.0025 mg	0.0100 mg
20 mg		0.0033 mg	0.0020 mg	0.0100 mg
20 mg *		-0.0005 mg	0.0020 mg	0.0100 mg
10 mg		0.0039 mg	0.0020 mg	0.0100 mg
5 mg		0.0036 mg	0.0020 mg	0.0100 mg
2 mg		0.0027 mg	0.0020 mg	0.0100 mg
2 mg *		0.0019 mg	0.0020 mg	0.0100 mg
1 mg		0.0034 mg	0.0020 mg	0.0100 mg

* Denotes weight is marked with a dot.

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Weight

ID Number	4000022685
Certificate Number	01061080-1
Date of Calibration	25-FEB-2019

NAME AND ADDRESS OF CUSTOMER

End user
Apex Instruments
204 Technology Park Ln
Fuquay Varina NC 27526

Client
Ohaus Corporation
1900 Polaris Parkway
Box D
Columbus OH 43240

SECTION 10: CALIBRATION PROCEDURE DATA

Nominal Mass Value	Serial Number	Standard Set No.	Cal Due	Balance Used	Cal Due	Procedure Used
20 g		MS002	08/01/19	A200XXL-132	01/01/20	Multi A-B
10 g		C006	08/01/19	A200XXL-133	01/01/20	Multi A-B
5 g		MS002	08/01/19	A200XXL-132	01/01/20	Multi A-B
2 g		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
2 g *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
1 g		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
500 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
200 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
200 mg *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
100 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
50 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
20 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
20 mg *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
10 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
5 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
2 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
2 mg *		A031	07/01/19	A5XL-131	01/01/20	Multi A-B
1 mg		A031	07/01/19	A5XL-131	01/01/20	Multi A-B

* Denotes weight is marked with a dot.

Balance Calibration

Make:	A&D Company Limited
Model:	HR-100AZ
SN:	6A7707793

	Weight (g)	Response (g)	% Difference
	0.0100	0.0099	-1.010%
	0.0200	0.0197	-1.523%
	0.0500	0.0499	-0.200%
	0.1000	0.0999	-0.100%
	0.2000	0.1996	-0.200%
	0.5000	0.4998	-0.040%
	1.0000	0.9999	-0.010%
	2.0000	1.9999	-0.005%
	5.0000	4.9997	-0.006%
	10.0000	9.9997	-0.003%
	20.0000	19.9992	-0.004%
	40.0000	39.9993	-0.002%
*	100.0000	100.0044	0.004%

* This balance is equipped with an internal 100g calibration weight for self calibration before every test. The 100g weight listed is a check weight provided with the balance.

Internally Calibrated: Yes

Date performed: 1/3/2025

Analyst: Brandon Jones

Calibration due: 1/3/2026

Weight Set SN:
4000022685

TARE WEIGHT LOG

Method 5 Filter Tare Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Samuel Burt	**Analyst:**	Samuel Burt	**Desiccator**	X
Date:	3/20/2024	**Date:**	3/22/2024	**Oven**	X
Time:	16:00	**Time:**	8:00	**Date Into Dryer:**	3/19/2024
Lab Temp, °F:	68	**Lab Temp, °F:**	68	**Time Into Dryer:**	3 hours
% RH:	35	**% RH:**	36		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
0.0500 g	0.0499	**0.1000 g**	0.1002	**Make**	A&D
0.1000 g	0.1000	**1.0000 g**	1.0000	**Model**	HR-100AZ
0.5000 g	0.4999			**Serial #**	6A7707793
1.0000 g	0.9998			**Calibration Weight Set SN:**	
3.0000 g	2.9994			4000022685	

Filter ID	Weight (g)	Weight (g)	Average (g)	Result
5B429	0.3522	0.3525	0.3524	Pass
5B430	0.3527	0.3529	0.3528	Pass
5B431	0.3523	0.3527	0.3525	Pass
5B432	0.3529	0.3530	0.3530	Pass

Completed By (Signature) / Date: 05/02/2024

Method 5 Beaker Tare Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	Brandon Jones	Analyst:	Brandon Jones	Desiccator	X
Date:	1/31/2025	Date:	1/31/2025	Oven	
Time:	8:35	Time:	16:05	Date Into Dryer:	1/30/2025
Lab Temp, °F:		Lab Temp, °F:		Time Into Dryer:	8:00
% RH:		% RH:			

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
10.0000 g	9.9999	**40.0000 g**	40.0000	Make	A&D
20.0000 g	20.0001	**100.0000 g**	100.0060	Model	HR-100AZ
30.0000 g	30.0001			Serial #	6A7707793
40.0000 g	40.0001			**Calibration Weight Set SN:**	
100.0000 g	100.0061			4000022685	

Beaker ID	Weight (g)	Weight (g)	Average (g)	Result
1	51.7664	51.7665	51.7665	Pass
2	51.8116	51.8116	51.8116	Pass
3	53.2784	53.2782	53.2783	Pass
Blank	51.9376	51.9372	51.9374	Pass

Completed By (Signature) / Date: Brandon Jones 1/30/2025

ANALYTICAL WEIGHT LOG

Method 5 Filter Analytical Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	BJ	**Analyst:**	BJ	**Desiccator**	X
Date:	2/3/2025	**Date:**	2/3/2025	**Oven**	
Time:	7:10	**Time:**	15:30	**Date Into Dryer:**	2/1/2025
Lab Temp, °F:	69	**Lab Temp, °F:**	73	**Time Into Dryer:**	All Day
% RH:	39	**% RH:**	41		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
0.0500 g	0.0500	**0.1000 g**	0.0999	**Make**	A&D
0.1000 g	0.0999	**1.0000 g**	1.0000	**Model**	HR-100AZ
0.5000 g	0.5001			**Serial #**	6A7707793
1.0000 g	0.9999			**Calibration Weight Set SN:**	
3.0000 g	3.0000			4000022685	

Filter ID	Sample ID	Weight (g)	Weight (g)	Average (g)	Result
SB429	R1	0.3526	0.3527	0.3527	Pass
SB430	R2	0.3530	0.3531	0.3531	Pass
SB431	R3	0.3533	0.3530	0.3532	Pass
SB432	FBlank	0.3534	0.3529	0.3532	Pass

Completed By (Signature) / Date: [signature] 2/3/2025

Method 5 Beaker Analytical Weight

Initial Weight Check		Final Weight Check		Drying Method	
Analyst:	BJ	**Analyst:**	BJ	**Desiccator**	X
Date:	2/3/2025	**Date:**	2/3/2025	**Oven**	
Time:	7:10	**Time:**	15:30	**Date Into Dryer:**	2/1/2025
Lab Temp, °F:	69	**Lab Temp, °F:**	73	**Time Into Dryer:**	All Day
% RH:	39	**% RH:**	41		

Cal Weight (g)	Weight (g)	Cal Weight (g)	Weight (g)	Balance Information	
10.0000 g	9.9999	**40.0000 g**	40.0001	**Make**	A&D
20.0000 g	20.0000	**100.0000 g**	100.0062	**Model**	HR-100AZ
30.0000 g	29.9999			**Serial #**	6A7707793
40.0000 g	40.0001			**Calibration Weight Set SN:**	
100.0000 g	100.0063			4000022685	

Beaker ID	Sample ID	Weight (g)	Weight (g)	Average (g)	Result
1	R1	51.7731	51.7732	51.7732	Pass
2	R2	51.8185	51.8183	51.8184	Pass
3	R3	53.2876	53.2877	53.2877	Pass
4	FBlank	51.9374	51.9375	51.9375	Pass

Completed By (Signature) / Date: 2/3/2025